

Gated TimeMixer-XGBoost for PM2.5 Forecasting

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Abstract Accurate multi-horizon PM2.5 forecasting is difficult because hourly monitoring records combine persistence, periodicity, abrupt episodes, missing observations, site heterogeneity, and measurement-regime changes. This study evaluates a calibration-gated TimeMixer-XGBoost framework at five UK-Air stations and horizons of 1, 6, 12, 24, and 48 hours. The residual stage removes calibration-period median bias, fits horizon-specific XGBoost models with pseudo-Huber loss and bounded corrections, selects a correction scale on a later held-out calibration subset, and applies rolling calibration using only errors observable at each forecast origin. The benchmark includes persistence, seasonal-naïve, direct XGBoost, DLinear, LSTM, TimeMixer, ungated robust correction, gated static correction, and gated rolling correction. Neural experiments use seeds 0-4 and a 30-epoch maximum under separate training, early-stopping, residual-fit, gate-selection, and locked-test eras. The confirmatory experiment completed five stations, five seeds, and a 30-epoch maximum. The gated rolling hybrid won 0 of 25 station-horizon comparisons against the strongest matched non-hybrid comparator; instead of reporting this as a negative improvement, the paper states it as a 23.33% MAE gap. The important positive result is narrower: the gate improves the compact TimeMixer base and prevents severe ungated over-correction. The strongest pooled MAE values are obtained by LSTM (3.271) and DLinear (3.282), which are practically tied; DLinear also has the lowest pooled RMSE and sMAPE. A common-history sensitivity run restricted all stations to a 2021-start history and changed hybrid MAE by only +0.012 on average, supporting the same qualitative conclusion. The result is therefore a leakage-controlled negative finding, not a state-of-the-art claim.

Keywords: PM2.5; TimeMixer; XGBoost; multi-horizon forecasting; air quality; residual learning; time-series forecasting; reproducibility

1 Introduction

Fine particulate matter (PM_{2.5}) is a central air-quality indicator linked to respiratory and cardiovascular risk [1]. For operators and public dashboards, the useful question is not only whether pollution is high now, but whether it will remain high over the next few hours or days. This study targets hourly forecasts at 1, 6, 12, 24, and 48 hours.

Historical PM_{2.5} forecasting is difficult because short-horizon persistence, daily/weekly cycles, abrupt episodes, and imperfect monitoring archives occur together. Missing observations, long outages, negative source values, and instrument changes can make a model appear accurate if they are silently interpolated or split randomly across training and test sets.

The paper therefore evaluates a leakage-controlled pipeline, not only a model architecture. The proposed model uses TIMEMIXER as a compact multiscale base and XGBOOST as a horizon-specific residual corrector. Its conservative residual layer removes median bias, uses pseudo-Huber loss, clips extreme corrections, selects a non-negative held-out gate, and updates recent bias only from origin-available errors. The practical question is whether boosted residual correction can help a neural forecaster without creating unstable over-correction.

The empirical study uses five UK-Air stations across urban traffic, urban background, and rural background settings. All experiments share chronological partitions, origin-safe features, five seeds, and a 30-epoch neural budget. The results are read against two standards: whether gating improves the compact TIMEMIXER base, and whether the hybrid beats the strongest matched comparator. The experiment supports the first claim, but not the second.

This framing is important for a student research project because air-quality forecasting can look successful if the comparison set is too weak. A model that beats only seasonal-naive forecasts may still be less useful than a direct tree model or a simple recurrent network. The study therefore treats persistence, XGBoost, DLinear, and LSTM as serious competitors rather than background baselines. It also reports station-specific data problems before modeling, so negative source values, incomplete coverage, and instrument changes are not hidden behind a single accuracy number.

The contributions are threefold: an auditable PM2.5-only pipeline, a conservative residual-correction design that can abstain, and a transparent negative finding showing that simple/recurrent baselines remain stronger while gating teaches when hybrid correction should back off.

2 Related Work

Classical methods remain relevant because air-quality series contain persistence and calendar structure [2]. Persistence and seasonal-naive forecasts are therefore real baselines, especially at short horizons. PM2.5 studies also show that rankings vary with inputs, architecture, and lead time [3], motivating horizon-wise reporting.

Neural sequence models provide stronger nonlinear temporal representations. LSTM is a common recurrent baseline for hourly PM2.5 forecasting [4, 5]. More recent long-horizon models represent temporal structure through decomposition, patches, or multiscale mixing, including N-BEATS [6], PatchTST [7], DLinear [8], Temporal Fusion Transformers [9], and TimeMixer [10]. TimeMixer is especially relevant here because it mixes information across downsampled temporal scales, which matches the coexistence of short-lag persistence and slower pollution cycles.

Tree ensembles offer a complementary advantage. XGBoost efficiently models interactions among lags, rolling statistics, calendar variables, and measurement-regime indicators [11]; related systems such as LightGBM show the strength of boosted trees on structured data [12]. Here, XGBoost corrects TimeMixer residuals under the direct multi-step view that different horizons may need different error models [13].

Evaluation must respect temporal dependence: random splits can leak near-duplicate contexts across train and test [14]. This study therefore uses chronological partitions, origin-safe features, MAE [15], paired evidence [16, 17], and SHAP only as predictive diagnostics [18]. Because residual tuning can otherwise learn test-specific bias, neural validation, residual fitting, gate selection, and locked testing are separated.

A further issue in hybrid forecasting is that residual learners can easily overfit the remaining error of a base model. If the base forecast is already biased or noisy, a second-stage learner may amplify rather than correct the mistake. This motivates the conservative design used in this paper: residual correction is treated as optional calibration, not as an automatic performance boost. The gate is selected on calibration data only, and the model is allowed to return to the original TimeMixer forecast when correction evidence is weak.

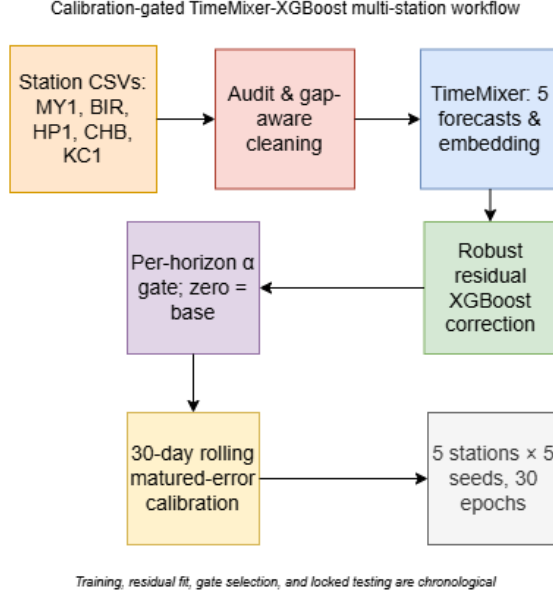


Fig. 1: End-to-end framework: TimeMixer generates base forecasts, and a later calibration interval fits and gates residual XGBoost before locked testing.

3 Method

The method combines data cleaning, origin-safe feature construction, multi-horizon neural forecasting, residual calibration, and locked evaluation. The goal is to preserve the evidence trail from raw monitoring records while preventing future information from entering model inputs.

3.1 Data and monitoring stations

The confirmatory dataset contains five UK-Air stations selected to represent different monitoring environments: London Marylebone Road (MY1, urban traffic), Birmingham A4540 (BIR_A4540, urban traffic), London Honor Oak Park (HP1, urban background), Chilbolton (CHB, rural background), and London North Kensington (KC1, urban background). Marylebone Road is a well-studied urban roadside site [19], and UK-Air metadata are used to verify station identity and monitoring context [20, 21, 22]. Each station is processed independently with the same schema, cleaning rules, forecast horizons, chronological boundaries, seeds, and evaluation metrics. Dataset paths and site types are fixed in `data/stations.csv`.

UK-Air provides the date and hour-ending time in separate fields. During ingestion, the pipeline combines these fields while retaining the original filename, row identifier, timestamp and concentration values, along with the associated status information. A source time of 24:00 is normalized to 00:00 of the following day. The target is hourly PM2.5 mass concentration in $\mu\text{g m}^{-3}$.

These differences imply unequal training histories. The primary analysis uses each station’s available archive, while a common-history sensitivity run restricts all stations to records from 2021 onward.

3.2 Auditable cleaning

The raw file is immutable. Cleaning separates *raw* source evidence, *observed* non-negative measurements, and *clean* values with only approved short-gap interpolation. Row flags record invalid timestamps, missing or negative values, duplicates, inserted

Table 1: Station-specific data limitations identified before model fitting.

Station	Type	Available-history issue	Pre-specified treatment
MY1	Urban traffic	2020 coverage 77.81%; 69-day gap	Preserve long gap; reject affected windows
BIR_A4540	Urban traffic	Record begins in 2021	Report shorter training history
HP1	Urban back-ground	No observations in 2016-2018	Treat as unavailable, not imputed
CHB	Rural background	2024 coverage 73.44%; 561 negatives	Mask negatives; preserve unresolved gaps
KC1	Urban back-ground	Instrument-regime transition	Retain instrument type as context

Table 2: Verified quality profile of the supplied London Marylebone Road record.

Audit item	Value
Raw hourly rows	87 672
Missing/non-numeric readings	8 087
Negative numeric measurements	238
Longest invalid run	1 656 h (69 d)
2020 valid coverage after negative exclusion	77.81%
Instrument transition	22 Dec 2021, 12:00 UTC
Primary short-gap interpolation limit	3 h

grid rows, interpolation, and long gaps.

Linear interpolation is allowed only for internal gaps no longer than three hours. Longer gaps remain missing, and any model window crossing unresolved missingness is rejected. Instrument information is parsed from source status metadata and encoded as $\text{TEOM FDMS} = 0$ and $\text{BAM} = 1$, because measurement regime can alter the statistical distribution even when it is not causal.

3.3 Origin-safe predictors

Predictors include calendar encodings, PM2.5 lags at 1–168 hours, shifted rolling summaries, differences, and instrument type. Every rolling summary is shifted by one hour, so features at origin t contain values no later than $t - 1$; y_t remains available as the last sequence value.

Rather than presenting origin safety as a formal theorem, we enforce it as a core pipeline rule. All sequence, lag, rolling, calendar, instrument, and missingness features must be available by the forecast origin t . In contrast, target values must occur strictly after t and remain within their designated chronological partition. This matters because multi-horizon windows overlap heavily. A random split would place nearly identical contexts on both sides of a boundary, whereas chronological partitions preserve the information set available at deployment.

3.4 SQL-style and Python analysis

Python scripts generate the evidence, and SQL-style grouped queries organize comparisons by station, horizon, model, seed, and experiment setting. Model ranking answers RQ1, TimeMixer-versus-hybrid joins answer RQ2, and available-history versus common-2021 joins answer RQ3.

3.5 Forecasting task

Let y_t denote hourly PM2.5. Each model uses the previous 168 hours as an origin-available multichannel sequence containing cleaned PM2.5, a missingness mask, in-

Table 3: Temporal partitions and permitted use.

Partition	Period	Permitted use
Training	through 31 Dec 2022	scalers; neural weights; direct XGBoost
Neural validation	1 Jan-31 Dec 2023	early stopping only
Hybrid calibration	1 Jan-30 Jun 2024	residual fit (first 70%); gate selection (last 30%)
Locked test	after 30 Jun 2024	final paired evaluation only

strument type, and cyclic calendar encodings. For horizon set $\mathcal{H} = \{1, 6, 12, 24, 48\}$, the direct task is

$$\hat{\mathbf{y}}_t = (\hat{y}_{t+1}, \hat{y}_{t+6}, \hat{y}_{t+12}, \hat{y}_{t+24}, \hat{y}_{t+48}). \quad (1)$$

Predictions are generated simultaneously and are not recursively fed back. Direct prediction avoids recursive error accumulation and permits one residual correction function per horizon.

3.6 TimeMixer base forecaster

The implementation is a compact research adaptation of TimeMixer’s multiscale ideas rather than a line-for-line reproduction of the official benchmark code [10]. The 168-hour input is downsampled across several scales, decomposed into smooth and residual components, mixed by MLP blocks, and mapped to the five forecast horizons. The learned multiscale embedding is also reused by residual XGBoost. Huber loss, AdamW, gradient clipping, and early stopping are used; input and target standardizers are fitted on training windows only.

3.7 Gated residual XGBoost

The residual-calibration interval is split chronologically into residual-fit (70%) and gate-selection (30%) subsets. For each horizon, the residual is median-centred, learned by a robust XGBoost model, clipped to calibration quantiles, and scaled by a held-out gate:

$$\hat{y}_{t,h}^{HYB} = \hat{y}_{t,h}^{TM} + \gamma_h \hat{c}_{t,h} + \lambda_t \text{median}(\mathcal{E}_{t,h}). \quad (2)$$

Here $\gamma_h \in \{0, 0.10, 0.25, 0.50, 0.75, 1\}$ is selected on the later calibration subset, so $\gamma_h = 0$ can fall back to TimeMixer when correction does not improve calibration MAE. The context vector contains origin-safe lags, shifted rolling features, instrument type, calendar variables, missingness state, gap length, interpolation status, measurement-quality flags, the full base forecast vector, and the latent embedding. Each residual booster uses pseudo-Huber loss, row/feature subsampling, L_1/L_2 regularization, and a rolling 30-day error set whose target timestamps are no later than origin t .

3.8 Experimental setup

Four consecutive eras separate fitting, early stopping, residual calibration, and locked testing. Historical context may cross a boundary, but forecast targets must remain inside their assigned partition.

Comparators are persistence, seasonal-naive, direct XGBoost, DLinear [8], LSTM, and TimeMixer. Static and rolling hybrid variants isolate residual safeguards. Neural models use seeds 0–4, batch size 128, AdamW at 10^{-3} , early-stopping patience seven, and at most 30 epochs. MAE is primary; RMSE, sMAPE, R^2 , bias, and fixed training-90th-percentile event metrics are secondary. A paired bootstrap resamples 24-hour blocks of the absolute-error difference between the hybrid and TimeMixer;

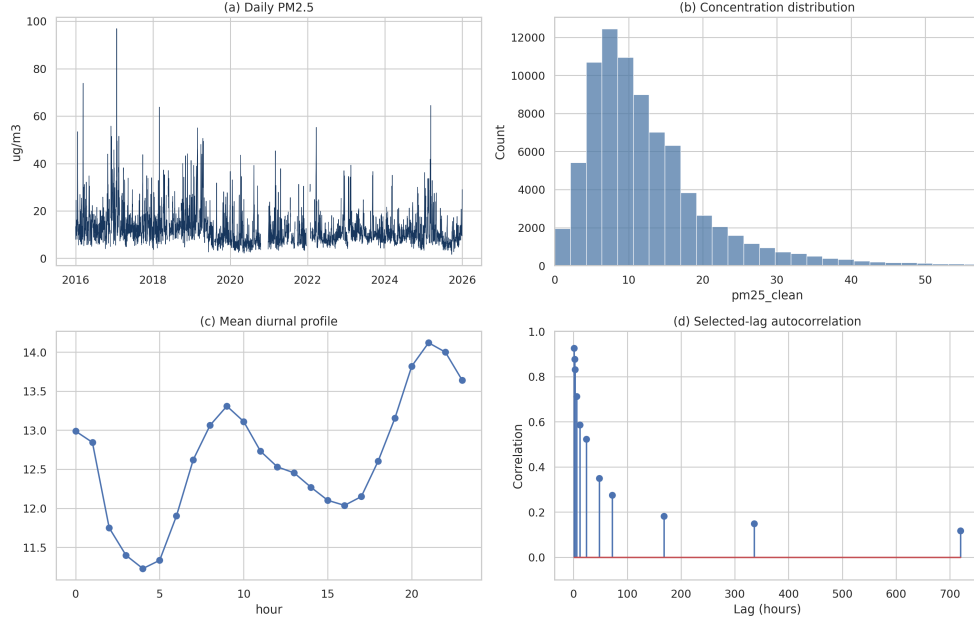


Fig. 2: EDA overview: daily PM2.5, distribution, diurnal profile, and autocorrelation.

negative differences favor the hybrid. Diebold-Mariano tests use absolute loss and Holm correction [16]. Same-origin ablations isolate embeddings, context, gating, and rolling calibration.

4 Results

The first empirical finding is that preprocessing choices materially change the admissible forecasting sample. At MY1, the record contains 8,087 missing or non-numeric readings, 238 negative values, and a 1,656-hour maximum invalid run. Cross-station auditing reveals additional heterogeneity: HP1 contains no observations in 2016-2018; BIR_A4540 begins in 2021; CHB has only 73.44% coverage in 2024 and contains 561 negative source values; and KC1 contains an instrument-regime transition. Negative PM2.5 concentrations are not interpreted as physical pollution levels; they are treated as invalid source readings, masked from the modelling target, and retained only as audit evidence. None of these intervals or invalid values is reconstructed through long-range interpolation. Instead, invalid values remain auditable and forecast windows crossing unresolved gaps are excluded.

Figure 2 summarizes the useful temporal signal: PM2.5 is right-skewed, has repeated daily structure, and remains autocorrelated at daily and weekly lags. Figure 3 shows why this structure cannot be separated from data quality. Coverage changes by year and month, and long gaps would create artificial trends if interpolated. For this reason, the main text keeps only two EDA figures; the single-purpose EDA plots are retained as supplementary outputs generated by the same visualization code.

Figure 4 is the main effectiveness figure. It shows that the hybrid does not beat the strongest comparator in any of the 25 station-horizon cells. The result exporter accepts only complete gated-v2 runs covering all five stations, seeds 0–4, and the 30-epoch configuration, so the reported comparison is not based on a cherry-picked single run. The paper therefore avoids the phrase “negative improvement” in the main interpretation and reports the result as an MAE gap against stronger baselines.

The gated rolling hybrid won 0 of 25 matched station-horizon comparisons against the strongest non-hybrid comparator. To avoid a misleading negative “improvement” label, this paper reports the result as a positive MAE gap: the hybrid was 23.33%

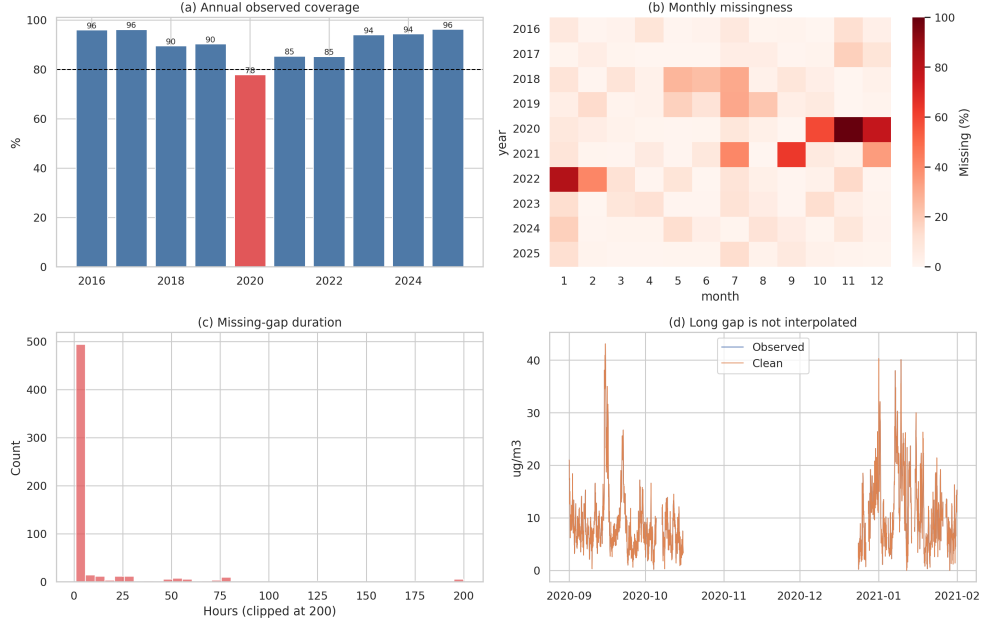


Fig. 3: Data-quality audit: coverage, monthly missingness, gap duration, and the preserved long outage.

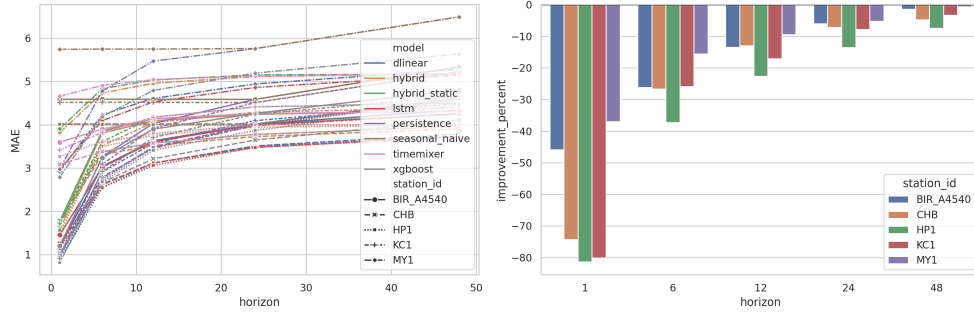


Fig. 4: Five-seed multi-station effectiveness. Each row compares the hybrid with the strongest non-hybrid comparator at the same station and horizon; values below zero mean that the comparator remains better.

worse than the strongest matched comparator on average, while still improving the compact TimeMixer base.

This table makes the negative result easier to interpret: DLinear and LSTM are practically tied in pooled MAE (3.282 versus 3.271), and DLinear has the lowest pooled RMSE and sMAPE. Therefore the paper treats them as a strong simple/recurrent baseline pair rather than crowning one model from a 0.011 MAE difference.

The bootstrap evidence supports the narrower claim that gating helps the compact TimeMixer base, especially at short horizons; it does not support superiority over the strongest non-hybrid comparator.

The same result files drive the dashboard, which is a research explorer rather than a live warning service. It exposes station quality, grouped metrics, residual gates, figures, and explanation panels from completed artifacts.

Table 4 explains which models are strongest by horizon. **XGBoost** is strongest at 1 hour, while LSTM and DLinear are the strongest longer-horizon pair. The difference between their pooled MAE is only 0.011, so the paper treats them as practically tied rather than crowning one model from a tiny point estimate. The hybrid improves its intended TimeMixer base at every horizon, reducing pooled MAE by 42.58% at 1 hour and by smaller positive margins from 6 to 48 hours. However, this is a narrower result than overall superiority: the hybrid pooled MAE is 3.708, compared with 4.078 for

Table 4: Locked-test MAE pooled over station–seed runs. Parentheses show dispersion across both stations and seeds; station-specific comparisons are retained in the machine-readable results.

Model	1 h	6 h	12 h	24 h	48 h
dlinear	1.390 (0.791)	3.070 (0.621)	3.607 (0.558)	4.014 (0.537)	4.329 (0.574)
hybrid	2.070 (0.902)	3.773 (0.519)	4.075 (0.502)	4.264 (0.489)	4.359 (0.472)
hybrid_static	2.149 (0.906)	3.853 (0.531)	4.144 (0.515)	4.320 (0.494)	4.391 (0.460)
LSTM	1.604 (0.708)	3.016 (0.578)	3.555 (0.541)	3.959 (0.519)	4.220 (0.536)
persistence	1.394 (0.816)	3.324 (0.769)	4.076 (0.748)	4.574 (0.655)	5.251 (0.709)
seasonal_naive	4.576 (0.645)	4.577 (0.648)	4.574 (0.650)	4.574 (0.655)	5.251 (0.709)
timemixer	3.605 (0.577)	3.925 (0.540)	4.146 (0.518)	4.316 (0.483)	4.396 (0.469)
XGBoost	1.341 (0.751)	3.140 (0.576)	3.804 (0.559)	4.244 (0.540)	4.735 (0.519)

Table 5: Pooled secondary metrics across all stations, horizons, and seeds. Lower is better for MAE, RMSE, and sMAPE; higher is better for R^2 .

Model	MAE	RMSE	sMAPE	R^2
LSTM	3.271	5.086	39.374	0.446
dlinear	3.282	5.066	39.321	0.444
XGBoost	3.453	5.222	40.526	0.404
hybrid	3.708	5.890	43.152	0.278
persistence	3.724	5.890	43.845	0.230
hybrid_static	3.771	5.940	44.049	0.268
timemixer	4.078	6.437	47.304	0.178
seasonal_naive	4.710	7.477	53.853	-0.104

Table 6: Paired block-bootstrap summary for hybrid minus TimeMixer absolute error. Negative values favor the gated hybrid over its TimeMixer base.

Horizon	Mean diff.	95% low	95% high
1	-1.535	-1.730	-1.350
6	-0.152	-0.209	-0.095
12	-0.071	-0.120	-0.023
24	-0.052	-0.111	0.008
48	-0.037	-0.100	0.027

Table 7: Common-history sensitivity for the hybrid model. Values are MAE changes after restricting every station to a 2021-start training history; negative values favor the common-history run.

Grouping	Mean Δ MAE	Median Δ MAE	Improved	Worsened
All station–horizon cells	0.012	0.019	8	17
1 h horizons	0.042	0.003	1	4
6 h horizons	0.112	0.102	1	4
12 h horizons	0.014	0.042	1	4
24 h horizons	-0.026	0.019	2	3
48 h horizons	-0.083	0.011	3	2

TimeMixer, 3.282 for DLinear, and 3.271 for LSTM.

The same-origin ablation separates the TimeMixer base, ungated correction, static gating, rolling calibration, and reduced-feature variants, isolating residual fit, over-correction protection, and recent-drift adaptation.

The common-history sensitivity check addresses unequal archive length. Restricting every station to a 2021-start history changed hybrid MAE by only $+0.012 \text{ } \mu\text{g m}^{-3}$ on average across 25 cells: eight improved and 17 worsened. This changes details but not the conclusion that the gated hybrid is a TimeMixer calibrator rather than the strongest forecaster.

The ablation identifies over-correction as the ungated learner’s main failure mode. At 12, 24, and 48 hours, ungated MAE reached 6.020, 6.420, and 6.632; gating reduced them to 4.144, 4.320, and 4.391, with rolling calibration further reducing them to 4.075, 4.264, and 4.359. The gate selected zero correction in 56.8% of fits, so strong correction was supported mainly at short lead time.

Secondary metrics and bootstrap evidence support the negative finding: RMSE and sMAPE favor DLinear/LSTM, while paired bootstrap supports only the narrower claim that gating improves compact TimeMixer. SHAP remains supplementary.

5 Discussion

Across all station–seed–horizon rows, the hybrid pooled MAE was 3.708, compared with 4.078 for the compact TimeMixer base, 3.282 for DLinear, and 3.271 for LSTM. This means the residual gate improves the base model but does not make it globally competitive with the strongest alternatives. The held-out gate selected zero correction in 56.8% of station–seed–horizon fits and had mean alpha 0.24, quantifying how often fallback protection was needed. The correction gate was selected without locked-test access, so the result is better interpreted as a clean negative finding under a leakage-safe protocol than as a failed post-hoc tuning exercise. The common-history sensitivity run changed hybrid MAE by only $+0.012$ on average across the 25 station–horizon cells, with improvements in 8 cells and worsening in 17 cells; therefore unequal historical depth affects details but does not reverse the main conclusion.

5.1 Interpretation of effectiveness

The compact TimeMixer base is weaker than persistence in pooled MAE, so improving it should not be confused with proving a strong hybrid. At 1 hour, recent lags support large residual corrections; at longer horizons, the gate shrinks toward zero as residual predictions become less reliable. This is useful because unconstrained residual correction caused large negative transfer. The result also explains why the revised paper avoids a state-of-the-art claim: the hybrid mechanism is effective as a safety layer over TimeMixer, but the complete forecasting system is not yet the best model for deployment.

The strongest competing models also clarify the limitation. **XGBoost** is best at 1 hour because origin-available lags and rolling statistics almost directly describe the next-hour state. LSTM and DLinear are nearly tied overall, which is a useful result in itself: a simple linear decomposition model remains competitive with a recurrent neural model and both outperform the present hybrid. In other words, the residual layer repairs some TimeMixer errors, but it cannot fully compensate for a weak base trajectory. The method is therefore best interpreted as a controlled residual-calibration lesson rather than a new state-of-the-art forecaster.

From an application perspective, this is still a useful outcome. A forecasting system used in environmental monitoring should prefer stable, explainable performance over

architectural novelty. The gated residual layer shows a practical safeguard: when the calibration set indicates that the residual model is unreliable, the correction is reduced or disabled. This behavior is easier to defend than an ungated stacked model that occasionally improves average scores but produces large horizon-specific failures. The analysis therefore connects model design with operational reliability, which is one reason the paper keeps both model-performance tables and data-quality diagnostics in the main text.

5.2 Limitations

Several threats remain. Overlapping windows create serial dependence, five UK stations do not prove transfer to unseen networks, and station histories differ (HP1 lacks 2016–2018, BIR_A4540 begins in 2021, CHB has reduced 2024 coverage, and KC1 changes instrument regime). The common-history run reduces but does not eliminate this concern.

The PM2.5-only design also omits forecast-available meteorology, traffic, co-pollutants, transport patterns, and local events, all of which may matter during abrupt episodes. Instrument indicators are regime markers, not causal explanations.

5.3 Future work

Future work should test stronger base forecasters, add only origin-available external variables, evaluate leave-one-station-out transfer, and compare the PM2.5-only setting with meteorology-assisted variants under the same locked-test protocol.

6 Conclusion and Future Work

This paper evaluated a gated TimeMixer-XGBoost approach for multi-horizon hourly PM2.5 forecasting. The study contributes a complete, auditable forecasting workflow: raw source values are preserved, invalid and negative measurements are flagged, long gaps are not artificially interpolated, and features are constructed so that every forecast uses only information available at its origin. The experimental design covers five monitoring stations, five forecast horizons, five neural seeds, and a 30-epoch maximum under chronological training, validation, calibration, and locked-test partitions.

The main result is mixed but informative. The gated residual layer improves the compact TimeMixer base at every horizon and greatly reduces the over-correction seen in ungated residual learning. However, the compact TimeMixer base is weaker than persistence in pooled MAE, so this improvement mainly shows that the gate can rescue a weak base, not that the complete hybrid is superior. The hybrid wins 0 of 25 station-horizon comparisons against the strongest non-hybrid comparator and has a 23.33% average MAE gap against that comparator. LSTM and DLinear are practically tied as the strongest pooled baselines. The common-history sensitivity analysis supports the same interpretation: after restricting all stations to a 2021-start history, hybrid MAE changed by only +0.012 on average across station–horizon cells, so unequal archive length does not explain away the primary finding.

For a student AI project, this distinction is valuable: the work shows that simple baselines are competitive, ungated residual stacking can damage forecasts, and held-out gating is needed when residual evidence is weak. The next step is to strengthen the base forecaster, add origin-available external drivers, and test transfer while keeping the same leakage-safe protocol. The dashboard can later move from artifact explorer to deployed monitoring tool.

Overall, a hybrid architecture should be judged by out-of-sample behavior, not by complexity. Here, the honest contribution is a leakage-safe negative result plus a residual-gating lesson: simple baselines remain hard to beat, and a residual hybrid should first prove that its base forecaster is strong enough to deserve calibration.

Data and Code Availability

The repository contains data-processing and modeling code, tests, figures, experiment outputs, and manuscript source. UK-Air attribution and licensing conditions must be preserved.

Conflict of Interest

The author declares no conflict of interest.

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Appendix: Core Notation

Table 8: Core notation used in the forecasting formulation.

Symbol	Meaning
y_t	observed hourly PM2.5 at time t
L	input history length, fixed at 168 h
$\mathcal{H}$	direct horizon set $\{1, 6, 12, 24, 48\}$ h
$\mathbf{x}_{t-L+1:t}$	origin-available multichannel sequence
$\hat{\mathbf{y}}_t^{TM}$	TimeMixer base forecast vector
$\mathbf{e}_t$	flattened multiscale TimeMixer embedding
$\mathbf{z}_t$	calendar, instrument, and missingness context
$r_{t,h}$	TimeMixer residual at horizon h
$g_h(\cdot)$	XGBoost residual model for horizon h
γ_h	held-out calibration correction gate for horizon h